PROJECT CHARTER

Project Title: AI Meeting Communication Coach: A Speaker-Aware Desktop Assistant

Project Sponsor: _____ **Date Prepared:** 16/08/2026

Project Manager: Sneha Neupane **Project Customer:** Professionals and Students looking to engage more effectively in meetings.

Executive Summary:

This project will build a desktop assistant that sits in a small floating window on top of Zoom, Teams or Meet. It captures microphone and system audio, transcribes the meeting live, and labels who is speaking.

It helps in two ways. During the meeting it shows key points, action items, topic changes, answers to questions just asked, and questions the user could ask next. It also flags questions that were asked but never answered. After the meeting it measures speaking time, turn taking and interruptions, then writes a short review for each speaker with suggestions for next time.

Meeting history is stored in a knowledge base built with Graphify so reviews can be compared over time. The prototype uses Flutter for macOS and Windows, Deepgram for transcription and diarisation, and Gemini for analysis. It runs for twelve weeks and is done by one student. Work has not started yet.

Project Purpose or Justification:

Two things go wrong in meetings. During the call, decisions and action items get lost and questions slide past unanswered. After the call, nobody is told how they took part, so quiet people stay quiet and interruptions go unnoticed.

Current tools only produce transcripts and summaries after the meeting, which is too late to help anyone. None of them measure how well people communicated, even though a speaker labelled transcript already holds that information.

This project covers both gaps in one application: live support while the meeting runs, and personal coaching afterwards. It also gives the student practical work in real time audio, speaker diarisation, LLM integration and cross platform desktop development.

Project Description:

A desktop prototype for macOS and Windows, built in Flutter. Audio capture and window control are written natively for each platform, everything else is shared code.

During the meeting

1. Capture microphone and system audio together.
2. Live transcription with speaker labels via Deepgram.
3. Live insights from Gemini: key points, action items, topic changes, answers to questions asked.
4. Suggested questions the user could ask next.
5. Alerts when a question goes unanswered.
6. Always on top overlay with compact and expanded modes, hotkeys, and hidden from screen sharing.

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7. Optional document upload (agenda, specification) used as context when that topic comes up.

After the meeting

8. Speaking time, turn taking and interruption counts per speaker.
9. A short written review for each speaker with one or two suggestions.
10. Results stored in a Graphify knowledge base so measures can be compared across meetings.

Out of scope: mobile and web, Linux, video or tone analysis, Zoom or calendar API integration, user accounts or cloud sync, languages other than English.

Assumptions: Deepgram diarisation is accurate enough for speaker level analysis; both APIs stay available at the expected cost; consented test meetings can be arranged; a macOS and a Windows machine are available.

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Initial Risks:

Risk Id	Risk Statement	Probability	Impact	Score	Strategy	Response
R1	Diarisation mislabels speakers on poor or overlapping audio	High	High		Mitigate	Test in Week 5, tune settings, fall back to single speaker mode
R2	Live insights arrive too late to be useful	High	High		Mitigate	Fixed analysis interval, short prompts, measure latency from Week 7
R3	Gemini output is wrong, invented or badly formatted	Medium	Medium		Mitigate	Structured JSON with fallback parsing, present output as suggestions only
R6	Windows audio capture takes longer than planned	Medium	Medium		Mitigate	One shared interface, macOS first, port in Week 9

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Initial Stakeholders:

Name	Interest	Impact	Strategies
Sneha Neupane (Student)	Grade, deadlines, learning	High	Owns all work, weekly progress log
Supervisor	Academic quality, feasible scope	High	Fortnightly meetings, agree scope changes in writing
Assessment panel	Fair marking of report and demo	High	Work mapped to marking criteria, demo rehearsed
Test participants	Consent and data handling	Medium	Written consent, local storage, deleted after evaluation
Deepgram and Google	Proper use of their APIs	Low	Stay within limits, keep keys out of the repository
University ethics process	Data protection compliance	Medium	Ethics approval before any recording

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Project Deliverables:

- Requirements and review of existing tools
- Architecture and interface design
- Overlay prototype with audio capture, transcription and diarisation (macOS and Windows)
- Live insight engine and unanswered question alerts
- Document upload with local relevance matching
- Participation and interruption analysis
- Per speaker review and feedback
- Graphify knowledge base with progress tracking
- Annotated test meetings and evaluation results
- Final report and demonstration

Summary Milestones	**Due Date**
Requirements complete	End of Week 2
Design signed off	End of Week 3
Overlay, capture, transcription, diarisation working	End of Week 5
Live insights and unanswered questions complete	End of Week 7
Per speaker review and progress tracking complete	End of Week 9
Evaluation complete	End of Week 11
Report and demonstration delivered	End of Week 12

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Summary Budget and Feasibility:

Hardware and development tools are already available and free. The only direct cost is API usage.

Item	**Cost (AUD)**
Deepgram transcription and diarisation	60 to 120
Gemini analysis and feedback	40 to 80
Hardware and tools	0
Total	**100 to 200**

The project is feasible because both APIs provide ready streaming and analysis endpoints, so the work is integration and analysis logic rather than model building. The main concern is the number of features, handled by the priority order in R4.

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Project Objectives and Success Criteria

Scope:

A working prototype on macOS and Windows that gives live insights during a meeting and per speaker reviews afterwards, with results stored for comparison across meetings. Both sides must be demonstrable on real test meetings.

Time:

All milestones met within one week of plan, report and demo submitted by the module deadline.

Cost:

Total spend stays within AUD 200.

Quality:

- Live insights appear within 15 seconds of the relevant point in the conversation.
- Unanswered question detection at least 75 percent accurate against annotated test meetings.
- Speaking time per speaker within 10 percent of manual measurement.
- Runs a 30 minute meeting without crashing on both platforms.

Other:

Acceptance Criteria:

1. Runs end to end on macOS and Windows with a live speaker labelled transcript.
2. Shows live insights, suggested questions and unanswered question alerts during a meeting.
3. Produces participation figures and a written review for each speaker afterwards.
4. Compares one speaker's measures across two or more meetings.
5. Evaluation results documented against the quality targets.
6. Report submitted by the deadline and accepted by the supervisor.

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Project Approval and Manager

Project Approval Requirements:

Approved when the supervisor accepts this charter and the scope. Final approval rests with the supervisor and assessment panel against the acceptance criteria and marking scheme. Scope, deliverable or schedule changes must be agreed with the supervisor in writing.

Project Manager:

Sneha Neupane (900008705), responsible for all planning, development, testing, documentation and reporting, and for raising problems with the supervisor early.

Approvals:
Sneha

Project Manager Signature	Sponsor or Originator Signature
Sneha Neupane	
Project Manager Name	Sponsor or Originator Name
16/08/2026	
Date	Date